list

In [4]:
```
l1 = []
l1
```

Out[4]:
```
[]
```

create a integer element list

In [5]:
```
l2 = [10,20,30,40]
l2
```

Out[5]:
```
[10, 20, 30, 40]
```

float elements

In [6]:
```
l3 = [10.0,20.2,30.3,40.4]
l3
```

Out[6]:
```
[10.0, 20.2, 30.3, 40.4]
```

string elements

In [7]:
```
l4 = ['one','two','three']
l4
```

Out[7]:
```
['one', 'two', 'three']
```

mixed data types

In [9]:
```
l5 = ['somu',24,40.5]
l5
```

Out[9]:
```
['somu', 24, 40.5]
```

In [10]:
```
l6 = [50,60,[10,20],[40,80]]
l6
```

Out[10]:
```
[50, 60, [10, 20], [40, 80]]
```

In [12]:
```
l7 = ['Asif', 25 ,[50, 100],[150, 90] , {'John' , 'David'}]
l7
```

Out[12]:
```
['Asif', 25, [50, 100], [150, 90], {'David', 'John'}]
```

In [13]:
```
len(l6)
```

Out[13]:
```
4
```

In [14]:
```
l2[0]
```

Out[14]: 10

In [15]:
```
l4[0]
```

Out[15]: 'one'

In [19]:
```
l5
```

Out[19]: ['somu', 24, 40.5]

In [20]:
```
l6
```

Out[20]: [50, 60, [10, 20], [40, 80]]

In [21]:
```
l6[2]
```

Out[21]: [10, 20]

In [22]:
```
l6[-1]
```

Out[22]: [40, 80]

In [23]:
```
my_list = ['one','two','three','four','five','six','seven','eight']
```

In [24]:
```
my_list[0:3]
```

Out[24]: ['one', 'two', 'three']

In [25]:
```
my_list[2:5]
```

Out[25]: ['three', 'four', 'five']

In [28]:
```
my_list[:3]
```

Out[28]: ['one', 'two', 'three']

In [30]:
```
my_list[:2]
```

Out[30]: ['one', 'two']

In [31]:
```
my_list[:-3]
```

Out[31]: ['one', 'two', 'three', 'four', 'five']

In [32]:
```
my_list[-3:]
```

Out[32]: ['six', 'seven', 'eight']

In [33]:
```
my_list[-2:]
```

Out[33]: ['seven', 'eight']

In [34]:
```
my_list[-1]
```

Out[34]: 'eight'

In [35]:
```
my_list[:]
```

Out[35]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']

In [38]:
```
my_list.append('nine')
my_list
```

Out[38]:
```
['one',
 'two',
 'three',
 'four',
 'five',
 'six',
 'seven',
 'eight',
 'nine',
 'nine',
 'nine']
```

In [40]:
```
my_list.remove('one')
my_list
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[40], line 1
----> 1 my_list.remove(     )
      2 my_list

ValueError: list.remove(x): x not in list
```

In [42]:
```
my_list.insert(1,'ONE')
my_list
```

Out[42]:
```
['two',
 'ONE',
 'three',
 'four',
 'five',
 'six',
 'seven',
 'eight',
 'nine',
 'nine',
 'nine']
```

In [45]:
```
my_list.pop()
```

Out[45]: 'nine'

In [46]:
```
my_list.pop(8)
my_list
```

Out[46]: ['two', 'ONE', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [47]:
```
my_list
```

Out[47]: ['two', 'ONE', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [48]:
```
my_list.remove('ONE')
```

In [49]:
```
my_list
```

Out[49]: ['two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [50]:
```
my_list.insert(0,'one')0
```

```
  Cell In[50], line 1
    my_list.insert(0,'one')0
                           ^
SyntaxError: invalid syntax
```

In [51]:
```
my_list.insert(0,'one')
```

In [52]:
```
my_list
```

Out[52]:
```
['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

In [54]:
```
my_list[0] = 1
my_list[1] = 2
my_list[2] = 3
my_list[3] = 4
```

In [55]:
```
my_list
```

Out[55]:
```
[1, 2, 3, 4, 'five', 'six', 'seven', 'eight', 'nine']
```

In [56]:
```
my_list.clear()
```

In [57]:
```
my_list
```

Out[57]:
```
[]
```

In [58]:
```
del my_list
my_list
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[58], line 2
      1 del my_list
----> 2 my_list

NameError: name 'my_list' is not defined
```

In [59]:
```
my_list = ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

In [60]:
```
mylist1 = my_list.copy()
mylist1
```

Out[60]:
```
['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

In [61]:
```
id(mylist1),id(my_list)
```

Out[61]:
```
(2039691772480, 2039691752768)
```

In [63]:
```
mylist1[0]
```

Out[63]:
```
'one'
```

In [65]:
```
mylist2 = mylist1.copy()
mylist2
```

Out[65]:
```
['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

In [66]:
```
mylist2[0] = 1
```

In [67]:
```
mylist2
```

Out[67]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']

In [68]:
```
list1 = [1,2,3,4]
list2 = [5,6,7,8]
```

In [69]:
```
list3 = list1 + list2
list3
```

Out[69]: [1, 2, 3, 4, 5, 6, 7, 8]

In [70]:
```
list2.extend(list1)
```

In [71]:
```
list2
```

Out[71]: [5, 6, 7, 8, 1, 2, 3, 4]

In [72]:
```
1 in list2
```

Out[72]: True

In [73]:
```
10 in list2
```

Out[73]: False

In [75]:
```
if 3 in list2:
   print("present in the list")
else :
   print( "not present in the list")
```

```
present in the list
```

In [76]:
```
list1
```

Out[76]: [1, 2, 3, 4]

In [77]:
```
list1.reverse()
```

In [78]:
```
list1
```

Out[78]: [4, 3, 2, 1]

In [79]:
```
list1
```

Out[79]: [4, 3, 2, 1]

In [80]:
```
list1 = list1[::-1]
list1
```

Out[80]: [1, 2, 3, 4]

In [81]:
```
list1
```

Out[81]: [1, 2, 3, 4]

In [82]:
```
list2
```

Out[82]: [5, 6, 7, 8, 1, 2, 3, 4]

In [83]:
```
list2.sort(reverse = True)
list2
```

Out[83]: [8, 7, 6, 5, 4, 3, 2, 1]

In [85]:
```
list2.sort()
```

In [86]:
```
list2.sort()
list2
```

Out[86]: [1, 2, 3, 4, 5, 6, 7, 8]

In [88]:
```
for i in enumerate(list2):
    print(i)
```

```
(0, 1)
(1, 2)
(2, 3)
(3, 4)
(4, 5)
(5, 6)
(6, 7)
(7, 8)
```

In [89]:
```
list2.count(1)
```

Out[89]: 1

In [90]:
```
list2.count(10)
```

Out[90]: 0

In [91]:
```
all(list2)
```

Out[91]: True

In [92]:
```
list2.insert(0,9)
list2
```

Out[92]: [9, 1, 2, 3, 4, 5, 6, 7, 8]

In [93]:
```
all(list2)
```

Out[93]: True

In [94]:
```
list2.append(0)
```

In [95]:
```
all(list2)
```

Out[95]: False

In [96]:
```
list2.pop()
```

Out[96]: 0

In [97]:
```
list2.insert(3,0)
```

In [98]:
```
list2
```

Out[98]: [9, 1, 2, 0, 3, 4, 5, 6, 7, 8]

In [99]:
```
all(list2)
```

Out[99]: False

In [ ]: